

Distributions

D.1

Solution:

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} &= -\frac{m_0 g}{\rho} \delta(x-a), \quad 0 < x < L, t > 0 \\ u(0, t) = u(L, t) &= 0, \quad t > 0 \\ u(x, t) = g(x), \quad \frac{\partial u(x, 0)}{\partial t} &= h(x), \quad 0 < x < L\end{aligned}$$

□

D.4

- (a) $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$
 (b) $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$

Solution:

- (a) Here we use $g(x)\delta(x) = g(0)\delta(x)$, and start by differentiating the expression $(g(x)\delta(x))$:

$$\begin{aligned}(g(x)\delta(x))' &= g'(x)\delta(x) + g(x)\delta'(x) \\ &\downarrow \\ g(x)\delta'(x) &= (g(x)\delta(x))' - g'(x)\delta(x) = \\ &= (g(0)\delta(x))' - g'(0)\delta(x) = g(0)\delta'(x) - g'(0)\delta(x)\end{aligned}$$

- (b) Using the result from (a) and differentiating $(g(x)\delta'(x))$:

$$\begin{aligned}
(g(x)\delta'(x))' &= g'(x)\delta'(x) + g(x)\delta''(x) \\
&\downarrow \\
g(x)\delta''(x) &= (g(x)\delta'(x))' - g'(x)\delta'(x) = \\
&= (g(0)\delta'(x) - g'(0)\delta(x))' - g'(x)\delta'(x) = \\
&= g(0)\delta''(x) - g'(0)\delta'(x) - g'(0)\delta'(x) + g''(0)\delta(x) = \\
&= g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)
\end{aligned}$$

□

D.5

- (a) $e^x \delta_1$
- (b) $x\delta'$
- (c) $\cos(x)\delta'$
- (d) $x^2\delta''$
- (e) $\int g(t)\delta(t-1) dt$

Solution:

(a)

$$e^x \delta_1 = e^x \delta_1 = e \delta_1$$

(b) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$

$$x\delta' = -\delta$$

(c) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$:

$$\cos(x)\delta' = \cos(0)\delta' + \sin(0)\delta = \delta'$$

(d) Using $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$:

$$x^2\delta'' = 2\delta$$

(e) Using $\int g(x)\delta(x-a) dx = g(a)$:

$$\int g(t)\delta(t-1)dt = g(1)$$

□

D.6

(a) $f(x) = e^{-x}\theta(x)$

(b) $f(x) = e^{-|x|}$

Solution:

(a) Using $\theta'(x) = \delta(x)$

$$f'(x) = (e^{-x}\theta(x))' = -e^{-x}\theta(x) + e^{-x}\delta(x) = \delta(x) - e^{-x}\theta(x)$$

$$f''(x) = (\delta(x) - e^{-x}\theta(x))' = \delta'(x) + e^{-x}\theta(x) - e^{-x}\delta(x) = \delta'(x) - \delta(x) + e^{-x}\theta(x)$$

(b) Expressing $|x|$ in distributions:

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases} = x\theta(x) - x(1 - \theta(x)) = 2x\theta(x) - x$$

$$f'(x) = (-2\theta(x) - 2x\delta(x) + 1)e^{-|x|} = (1 - 2\theta(x))e^{-|x|}$$

$$\begin{aligned} f''(x) &= (e^{-|x|} - 2e^{-|x|}\theta(x))' = (1 - 2\theta(x))e^{-|x|} + 2(1 - 2\theta(x))e^{-|x|}\theta(x) - 2e^{-|x|}\delta(x) = \\ &= e^{-|x|}[(1 - 2\theta(x)) + 2\theta(x)(1 - 2\theta(x))] - 2\delta(x) = e^{-|x|}(1 - 2\theta(x))^2 - 2\delta(x) = \\ &= [(1 - 2\theta(x))^2 = 1 \text{ for all } x \neq 0] = e^{-|x|} - 2\delta(x) \end{aligned}$$

□

D.8

(b) Determine all primitive distributions for $x\theta(x-1)$

Solution:

(a)

$$(x^2\theta(x-1))' = 2x\theta(x-1) + x^2\theta'(x-1) = 2x\theta(x-1) + \delta(x-1)$$

(b) Try to compensate for the $\delta(x-1)$ in (a) by $x^2 \rightarrow x^2 - 1$:

$$((x^2 - 1)\theta(x-1))' = 2x\theta(x-1) + (x^2 - 1)\delta(x-1) = 2x\theta(x-1)$$

Now compensate with a factor 2 and we get the solution

$$\left(\frac{x^2 - 1}{2}\right)\theta(x-1) + C$$

□

D.9

Solution:

$$\frac{\partial^2 u}{\partial t^2} - S \frac{\partial^2 u}{\partial x^2} = F\delta(x-a)$$

$$u(0, t) = u(1, t) = 0$$

Since we are looking for the stationary solution, $\frac{\partial^2 u}{\partial t^2} = 0$ and we have

$$-\frac{\partial^2 u}{\partial x^2} = \frac{F}{S}\delta(x-a)$$

Integrating both sides twice gives

$$-u(x, t) = \frac{F}{S}(x-a)\theta(x-a) + Ax + B$$

Now check the boundary conditions for this expression:

$$u(0, t) = \frac{F}{S}(-a)\theta(-a) + B = 0 \rightarrow B = 0$$

$\underbrace{\hspace{1.5cm}}_{=0 \text{ since } a>0}$

$$u(1, t) = \frac{F}{S}(a-1)\theta(1-a) + A = 0 \rightarrow A = \frac{F}{S}(1-a)$$

$$u(x, t) = \frac{F}{S}((x(1-a) + (a-x)\theta(x-a)))$$

□

D.10**Solution:**

□

D.12

$$u'(x) + (2x + 1)u(x) = \delta'(x)$$

Solution:

Integrating factor for LHS is

$$e^{x^2+x}$$

Multiply this into the equation and integrate LHS, then using the expansion for $f(x)\delta'(x)$:

$$\begin{aligned} \left(u(x)e^{x^2+x}\right)' &= e^{x^2+x}\delta'(x) = e^0\delta'(x) - (2x+1)_{x=0}\delta(x) = \\ &= \left(u(x)e^{x^2+x}\right)' = \delta'(x) - \delta(x) \end{aligned}$$

Now integrating both sides gives:

$$u(x)e^{x^2+x} = \delta(x) - \theta(x) + C$$

$$\downarrow$$

$$u(x) = e^{-(x^2+x)}(\delta(x) - \theta(x) + C) = \delta(x) + e^{-x^2-x}(C - \theta(x))$$

□

D.15

- (a) $xU = 0$
- (b) $(x-1)U = 0$
- (c) $(e^x - 2)U = 0$
- (d) $(x^2 - 1)U = 0$
- (e) $\sin(x)U = 0$
- (f) $x^3U = 0$

Solution:

(a)

$$U = \delta(x) \rightarrow x\delta(x) = 0 \rightarrow U = c\delta(x), \quad c = \text{constant}$$

(b) Same as in (a) but $\delta(x - 1)$

$$U = c\delta(x - 1)$$

(c) We can try to set U as $\delta(x - a)$ to make $(e^x - 2) = 0 \rightarrow e^x = 2 \rightarrow x = \ln(2)$

$$U = c\delta(x - \ln(2))$$

(d) Same idea as above, but here we have two roots, $x = 1, x = -1$

$$U = c_1\delta(x - 1) + c_2\delta(x + 1)$$

(e) Can use $U = \delta(x - a)$ again to make $\sin(x) = 0$

$$U = c_k\delta(x - k\pi) = \sum_{k=-\infty}^{\infty} c_k\delta(x - k\pi)$$

(f) Triple root at $x = 0$, use the relation $x^k U = 0 \rightarrow U = \sum_{k=0}^{n-1} c_k \delta^{(k)}(x)$

$$U = c_0\delta(x) + c_1\delta'(x) + c_2\delta''(x)$$

□

D.17(a) $\theta(-2x + 1)$ (b) $\delta(2x)$ (c) $\delta(-2x + 1)$ (d) $\delta'(2x)$ **Solution:**

(a)

$$\theta(-2x+1) = \begin{cases} 1, & -2x+1 > 0 \rightarrow x < \frac{1}{2} \\ 0, & -2x+1 < 0 \rightarrow x > \frac{1}{2} \end{cases}$$

$$\downarrow$$

$$\theta(-2x+1) = \theta\left(-2\left(x - \frac{1}{2}\right)\right) = 1 - \theta\left(x - \frac{1}{2}\right) = 1 - \theta_{\frac{1}{2}}(x)$$

(b) Scaling rule gives

$$\delta(2x) = \frac{1}{2}\delta(x)$$

(c)

$$\delta(-x) = \delta(x), \delta(ax) = \frac{1}{|a|}\delta(x)$$

$$\downarrow$$

$$\delta(-2x+1) = \delta\left(-2\left(x - \frac{1}{2}\right)\right) = \frac{1}{2}\delta\left(x - \frac{1}{2}\right) = \frac{1}{2}\delta_{\frac{1}{2}}(x)$$

(d)

$$\delta'(ax) = \frac{1}{a|a|}\delta'(x)$$

$$\downarrow$$

$$\delta'(2x) = \frac{1}{4}\delta'(x)$$

□

D.18

(a) $\delta * e^{-x^2}$

(b) $\delta' * (e^{-x}\theta(x))$

(c) $\delta_1 * \sin(2x)$

Solution:

Using the definition of the convolution

$$\varphi * U = \int \varphi(x-y)U(y) \, \mathrm{d}$$

(a)

$$\int e^{-(x-y)^2} \delta(y) \, dy = e^{-x^2}$$

(b)

$$\begin{aligned} \int e^{-(x-y)} \theta(x-y) \delta'(y) \, dy &= [e^{y-x} \theta(x-y) \delta(y)]_{-\infty}^{\infty} - \int (e^{y-x} \theta(x-y))' \delta(y) \, dy = \\ &= (e^{-x} \theta(x))' = -e^{-x} \theta(x) + e^{-x} \delta(x) = \delta(x) - e^{-x} \theta(x) \end{aligned}$$

(c)

$$\int \sin(2(x-y)) \delta(y-1) \, dy = \sin(2x-2)$$

□

D.22

Solution:

Definition of the cosine series expansion for distributions

$$\begin{aligned} U &= \frac{c_0}{2} + \sum_{k=1}^{\infty} c_k \cos\left(\frac{k\pi x}{L}\right), \quad 0 < x < L \\ c_k &= \frac{2}{L} \int_0^L U(x) \cos\left(\frac{k\pi x}{L}\right) \, dx \end{aligned}$$

We will also use the expansion of $f(x)\delta'(x) = f(0)\delta'(x) - f'(0)\delta(x)$.

This gives ($k \neq 0$)

$$\begin{aligned} c_k &= \frac{2}{\pi} \int_0^{\pi} \delta'\left(x - \frac{\pi}{2}\right) \cos(kx) \, dx = \frac{2}{\pi} \int_0^{\pi} \cos\left(k\frac{\pi}{2}\right) \delta'\left(x - \frac{\pi}{2}\right) + k \sin\left(k\frac{\pi}{2}\right) \delta\left(x - \frac{\pi}{2}\right) \, dx = \\ &= \frac{2}{\pi} \int_0^{\pi} k \sin\left(\frac{k\pi}{2}\right) \, dx = \frac{2}{\pi} k \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

and for $k = 0$:

$$c_0 = \frac{2}{\pi} \int_0^{\pi} \delta' \left(x - \frac{\pi}{2} \right) dx = \frac{2}{\pi} \left[\delta \left(x - \frac{\pi}{2} \right) \right]_0^{\pi} = 0$$

$$\delta' \left(x - \frac{\pi}{2} \right) = \frac{2}{\pi} \sum_{k=1}^{\infty} k \sin \left(\frac{k\pi}{2} \right) \cos(kx)$$

□

D.23

Solution:

Whut □

D.24

Solution:

Diffusion,

$$\begin{aligned} \frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < L, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(L, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= M \delta_{\frac{L}{2}}, & 0 < x < L \end{aligned}$$

We have homogenous Neumann boundary conditions, which should give us a solution in the form of a cosine series expansion.

$$u(x, t) = \sum u_k(t) \cos \left(\frac{k\pi x}{L} \right)$$

Differentiating termwise and putting into the differential equation gives

$$\sum (u'_k(t) + \alpha u_k(t)) \cos\left(\frac{k\pi x}{L}\right) = 0, \quad \alpha = \frac{Dk^2\pi^2}{L^2}$$

$$\downarrow$$

$$u'_k(t) + \alpha u_k(t) = 0$$

$$\downarrow$$

$$u_k(t) = c_k e^{-\alpha t}$$

We can also expand $u(x, 0) = M\delta_{\frac{L}{2}}$ in a cosine series, to get the coefficients, $u_k(0) = c_k$

$$u(x, 0) = M\delta_{\frac{L}{2}} = \sum a_k \cos\left(\frac{k\pi x}{L}\right)$$

$$c_0 = \frac{1}{L} \int_0^L M\delta_{\frac{L}{2}} dx = \frac{M}{L}$$

$$c_k = \frac{2}{L} \int_0^L M\delta_{\frac{L}{2}} \cos\left(\frac{k\pi x}{L}\right) dx = \frac{2M}{L} \cos\left(\frac{k\pi}{2}\right) = , \quad k = 1, 2, 3, \dots$$

This gives the full solution

$$u(x, t) = \frac{M}{L} \left(1 + 2 \sum_{k=1}^{\infty} \cos\left(\frac{k\pi}{2}\right) e^{-\alpha t} \cos\left(\frac{k\pi x}{L}\right) \right)$$

□

D.25

Solution:

□

D.27

Solve the three-dimensional potential problem, with $r = \sqrt{x^2 + y^2 + z^2}$

$$\Delta u = \delta(r - 2), \quad 1 < r < 3$$

$$u = 1, \quad r = 1$$

$$u = 3, \quad r = 3$$

Solution:

Symmetric in r so the Laplacian becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \delta(r-2)$$

$$\left(r^2 \frac{\partial u}{\partial r} \right)' = r^2 \delta(r-2) = 4\delta(r-2)$$

↓

$$r^2 \frac{\partial u}{\partial r} = 4\theta(r-2) + A$$

$$\frac{\partial u}{\partial r} = \frac{4}{r^2} \theta(r-2) + \frac{A}{r^2}$$

$$u(r) = \left(2 - \frac{4}{r} \right) \theta(r-2) - \frac{A}{r} + B$$

$$u(1) = 1 = -\theta(-1) - A + B = 0 \rightarrow -A + B = 1$$

$$u(3) = 3 = \left(\frac{2}{3} \right) \theta(1) - \frac{A}{3} + B = \frac{2}{3} + \frac{2}{3}A = 3$$

↓

$$\begin{cases} -A + B = 1 \\ \frac{2}{3} - \frac{A}{3} + B = 3 \end{cases} \rightarrow \frac{2}{3} + \frac{2}{3}A = 2 \rightarrow A = 2, B = 3$$

↓

$$u(r) = \left(2 - \frac{4}{r} \right) \theta(r-2) - \frac{2}{r} + 3$$

□

D.29**Solution:**

$$-\Delta u = -u + \delta(x-a)\delta(y-b), \quad 0 < x < 2, 0 < y < 1$$

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -u + \delta_{a,b}$$

□

D.30

- (a) $\delta(t-1)$
- (b) $\delta'(t+2)$
- (c) 2
- (d) $\sin(2t)$
- (e) $(t+1)^2$
- (f) $e^t\theta(t)$

Solution:

(a) The Fourier transform of $\delta(x)$ is $\mathcal{F}[\delta(x)] = 1$ and the translation rule is $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} \mathcal{F}[f(t)]$

$$\mathcal{F}[\delta(t-1)] = e^{-i\omega}$$

(b) The derivative rule is $\mathcal{F}[f'(t)] = i\omega \mathcal{F}[f(t)]$. Using this and the translation rule

$$\mathcal{F}[\delta'(t+2)] = i\omega e^{2i\omega}$$

(c) Fourier transform of a constant is $2\pi\delta(\omega)$

$$\mathcal{F}[2] = 4\pi\delta(\omega)$$

(d)

$$\sin(2t) = \frac{e^{2it} - e^{-2it}}{2i}$$

We have $\mathcal{F}[e^{i\omega_0 t} f(t)] = \mathcal{F}[f(\omega - \omega_0)]$

$$\mathcal{F}\left[\frac{e^{2it}}{2i} \cdot 1 - \frac{e^{-2it}}{2i} \cdot 1\right] = \frac{\pi}{i}(\delta(\omega-2) - \delta(\omega+2))$$

(e) $\mathcal{F}[tf(t)] = i \frac{d}{d\omega} \mathcal{F}[f(t)]$

$$\begin{aligned} \mathcal{F}[(t+1)^2] &= \mathcal{F}[t^2 + 2t + 1] = i^2 \frac{d^2}{d\omega^2} 2\pi\delta(\omega) + i \frac{d}{d\omega} 4\pi\delta(\omega) + 2\pi\delta(\omega) = \\ &= -2\pi\delta''(\omega) + 4\pi i\delta'(\omega) + 2\pi\delta(\omega) \end{aligned}$$

(f) This distribution is not tempered, and thus cannot be Fourier transformed. $\square$

D.31

- (a) $\delta(\omega-1)$
- (b) $\delta'(\omega+2)$

(c) 2

(d) $\sin(2\omega)$ (e) $(\omega + 1)^2$ **Solution:**

Going backwards on the identities gives

(a)

$$\begin{aligned}
\mathcal{F}[1] &= 2\pi\delta(\omega) \\
\mathcal{F}[e^{i\omega_0 t} \cdot 1] &= 2\pi\delta(\omega - \omega_0) \\
&\downarrow \\
\mathcal{F}^{-1}[\delta(\omega - 1)] &= \frac{e^{it}}{2\pi}
\end{aligned}$$

(b)

$$\begin{aligned}
\mathcal{F}[t] &= 2\pi i\delta'(\omega) \\
\mathcal{F}[e^{i\omega_0 t} t] &= 2\pi i\delta'(\omega - \omega_0) \\
&\downarrow \\
\mathcal{F}^{-1}[\delta'(\omega + 2)] &= \frac{e^{-2it}}{2\pi i} t
\end{aligned}$$

(c)

$$\begin{aligned}
\mathcal{F}[\delta(t)] &= 1 \\
&\downarrow \\
\mathcal{F}^{-1}[2] &= 2\delta(t)
\end{aligned}$$

(d)

$$\begin{aligned}
\mathcal{F}[\theta(t+1) - \theta(t-1)] &= 2\frac{\sin(\omega)}{\omega} \\
\mathcal{F}\left[\theta\left(\frac{t}{2} + 1\right) - \theta\left(\frac{t}{2} - 1\right)\right] &= \mathcal{F}[\theta(t+2) - \theta(t-2)] = 2\frac{\sin(2\omega)}{\omega} \\
\mathcal{F}[(\theta(t+2) - \theta(t-2))'] &= \mathcal{F}[\delta(t+2) - \delta(t-2)] = i\omega\left(2\frac{\sin(2\omega)}{\omega}\right) = 2i\sin(2\omega) \\
&\downarrow \\
\mathcal{F}^{-1}[\sin(2\omega)] &= \frac{1}{2i}(\delta(t+2) - \delta(t-2))
\end{aligned}$$

(e)

$$\begin{aligned}
\mathcal{F} [\delta'(t)] &= i\omega \\
\mathcal{F} [-\delta''(t)] &= \omega^2 \\
\mathcal{F} [\delta(t)] &= 1 \\
&\downarrow \\
\mathcal{F}^{-1} [\omega^2 + 2\omega + 1] &= -\delta''(t) - 2i\delta'(t) + \delta(t)
\end{aligned}$$

□

D.34**Solution:**

□

D.36

- (a) $\delta(t-1)$
- (b) $\delta'(t+2)$
- (c) 2
- (d) $\sin(2t)$
- (e) $(t+1)^2$
- (f) $e^t\theta(t)$

Solution:

$$\begin{aligned}
\mathcal{L} [\delta(t)] &= 1 \\
\mathcal{L} [f(t-t_0)] &= e^{-t_0s} \mathcal{L} [f(t)] \\
\mathcal{L} [f'(t)] &= s\mathcal{L} [f(t)] \\
\mathcal{L} [e^{at}f(t)] &= \mathcal{L} [f(t-a)] \\
\mathcal{L} [\theta(t)] &= \frac{1}{s}
\end{aligned}$$

(a)

$$\mathcal{L} [\delta(t-1)] = e^{-s}$$

(b)

$$\mathcal{L} [\delta'(t+1)] = se^{2s}$$

(c) Not possible

(d) Not possible

(e) Not possible

$$\mathcal{L} [(t+1)^2] = \mathcal{L} [t^2 + 2t + 1]$$

(f)

$$\mathcal{L} [e^t \theta(t)] = \frac{1}{s-1}, \quad \operatorname{Re} s > 0$$

□

D.41

Expand the differential equations to $\mathbb{R}$ so that the boundary conditions are fulfilled.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u(0) = 1 \end{cases}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u'(0) = 1 \end{cases}$$

Solution:

Dirichlet conditions give odd expansion, Neumann conditions give even expansions.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u(0) = 1 \end{cases} \rightarrow (u^-)'' - iu^- = 2\delta', \quad x \in \mathbb{R}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u'(0) = 1 \end{cases} \rightarrow (u^+)'' - iu^+ = 2\delta, \quad x \in \mathbb{R}$$

□